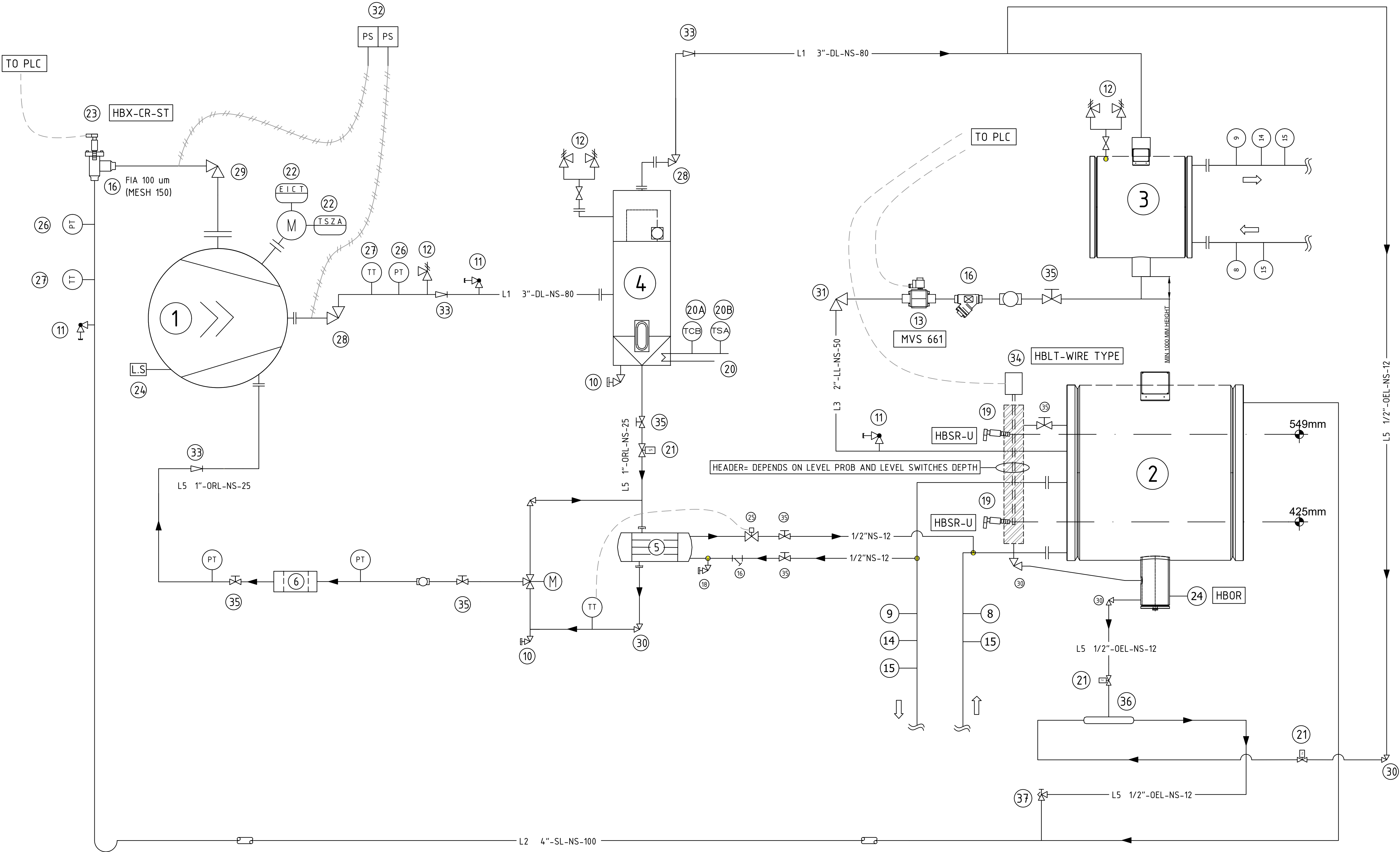


	EQUIPMENT LIST
①	COMPRESSOR
②	FLOODED EVAPORATOR (SHELL & PLATE)
③	CONDENSER (SHELL & PLATE)
④	OIL SEPARATOR
⑤	OIL COOLER
⑥	OIL FILTER
⑦	WATER INLET TEMPERATURE SENSOR
⑧	WATER OUTLET TEMPERATURE SENSOR
⑨	CRANKCASE HEATER (BUILT-IN)
⑩	MANUAL OIL DRAIN VALVE
⑪	SERVICE VALVE/ PURGE VALVE
⑫	SAFETY VALVE HIGH PRESSURE SIDE
⑬	SAFETY VALVE LOW PRESSURE SIDE
⑭	ELECTRONIC EXPANSION VALVE
⑮	FLOW SWITCH WATER SIDE
⑯	PRESSURE TRANSDUCER WATER SIDE
⑰	STRAINER
⑱	CHECK VALVE
⑲	DRAIN VALVE FOR WATER SIDE
⑳	LIQUID LEVEL SENSOR
㉑	OIL HEATER WITH THERMOSTAT (20A) AND TEMPERATURE LIMITER (20B)
㉒	SOLENOID VALVE
㉓	RATED CURRENT LIMITER & THERMAL WINDING PROTECTION-COMPRESSOR DRIVING MOTOR
㉔	VAPOR QUALITY SENSOR (HB)
㉕	OIL LEVEL SENSOR (HB)
㉖	TWO WAY CONTROL VALVE (MODULATING)
㉗	PRESSURE SENSOR
㉘	TEMPERATURE SENSOR
㉙	DISCHARGE ANGLE STOP VALVE
㉚	SUCTION ANGLE STOP VALVE
㉛	OIL SIDE ANGLE STOP VALVE
㉜	LIQUID LINE ANGLE STOP VALVE
㉝	DUAL PRESSURE SWITCH
㉞	CHECK VALVE
㉟	LIQUID LEVEL PROBE(HB)
㊱	STRAIGHT STOP VALVE
㊲	OIL EJECTOR
㊳	OIL CONTROL VALVE
L1	DISCHARGE LINE
L2	SUCTION LINE
L3	LIQUID LINE
L4	LIQUID INJECTION LINE
L5	OIL RETURN LINE

SYMBOL & LEGENDS	
	COMPRESSOR
	PURGE VALVE/SERVICE VALVE
	PIPE INSULATION
	TEMPERATURE TRANSDUCER
	PRESSURE TRANSDUCER
	BALL VALVE (STRAIGHT)
	BALL VALVE (ANGLE)
	SOLENOID VALVE
	PRESSURE SWITCH
	PRV (PRESSURE RELIEF VALVE)
	CHECK VALVE/NON RETURN VALVE
	SIGHT GLASS
	AIR VENT (WATER SIDE)
	3 WAY VALVE
	FLOW SWITCH



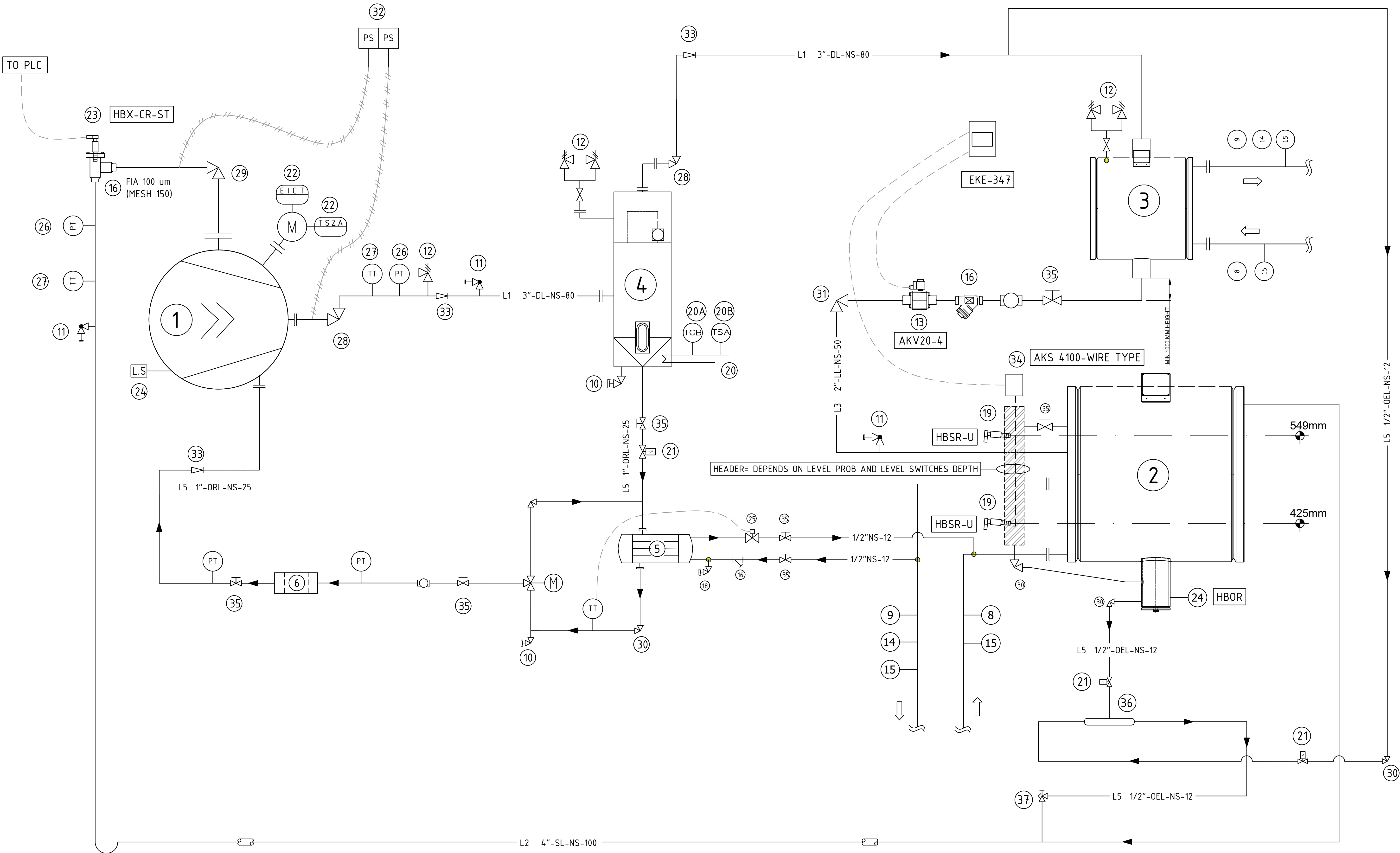
- TRAP AT THE BOTTOM OF RISER
- PREVENT LIQUID AMMONIA FROM RUNNING BACK INTO THE EVAPORATOR.
 - ENSURE PROPER OIL RETURN TO COMPRESSOR WITH GAS VELOCITY.

PIPING AND INSTRUMENTATION DIAGRAM

PROJECT NAME:	690KW HEAT PUMP(R717)	 HEATING + COOLING
PROJECT NUM:	2506-19	
REV:	9	
DATE:	25-SEP-25	
		WATER TO WATER HEAT PUMP-690KW (R717)
		REFRIGERATION PIPING AND INSTRUMENTATION DIAGRAM

	EQUIPMENT LIST
1	COMPRESSOR
2	FLOODED EVAPORATOR (SHELL & PLATE)
3	CONDENSER (SHELL & PLATE))
4	OIL SEPARATOR
5	OIL COOLER
6	OIL FILTER
7	WATER INLET TEMPERATURE SENSOR
8	WATER OUTLET TEMPERATURE SENSOR
9	CRANKCASE HEATER (BUILT-IN)
10	MANUAL OIL DRAIN VALVE
11	SERVICE VALVE/ PURGE VALVE
12	SAFETY VALVE HIGH PRESSURE SIDE
12A	SAFETY VALVE LOW PRESSURE SIDE
13	ELECTRONIC EXPANSION VALVE
14	FLOW SWITCH WATER SIDE
15	PRESSURE TRANSDUCER WATER SIDE
16	STRAINER
17	CHECK VALVE
18	DRAIN VALVE FOR WATER SIDE
19	LIQUID LEVEL SENSOR
20	OIL HEATER WITH THERMOSTAT (20A) AND TEMPERATURE LIMITER (20B)
21	SOLENOID VALVE
22	RATED CURRENT LIMITER & THERMAL WINDING PROTECTION-COMPRESSOR DRIVING MOTOR
23	VAPOR QUALITY SENSOR (HB)
24	OIL LEVEL SENSOR (HB)
25	TWO WAY CONTROL VALVE (MODULATING)
26	PRESSURE SENSOR
27	TEMPERATURE SENSOR
28	DISCHARGE ANGLE STOP VALVE
29	SUCTION ANGLE STOP VALVE
30	OIL SIDE ANGLE STOP VALVE
31	LIQUID LINE ANGLE STOP VALVE
32	DUAL PRESSURE SWITCH
33	CHECK VALVE
34	LIQUID LEVEL PROBE(HB)
35	STRAIGHT STOP VALVE
36	OIL EJECTOR
37	OIL CONTROL VALVE
L1	DISCHARGE LINE
L2	SUCTION LINE
L3	LIQUID LINE
L4	LIQUID INJECTION LINE
L5	OIL RETURN LINE

SYMBOL & LEGENDS	
	COMPRESSOR
	PURGE VALVE/SERVICE VALVE
	PIPE INSULATION
	TEMPERATURE TRANSDUCER
	PRESSURE TRANSDUCER
	BALL VALVE (STRAIGHT)
	BALL VALVE (ANGLE)
	SOLENOID VALVE
	PRESSURE SWITCH
	PRV (PRESSURE RELIEF VALVE)
	CHECK VALVE/NON RETURN VALVE
	SIGHT GLASS
	AIR VENT (WATER SIDE)
	3 WAY VALVE
	FLOW SWITCH



TRAP AT THE BOTTOM OF RISER

- PREVENT LIQUID AMMONIA FROM RUNNING BACK INTO THE EVAPORATOR.
- ENSURE PROPER OIL RETURN TO COMPRESSOR WITH GAS VELOCITY.

PIPING AND INSTRUMENTATION DIAGRAM

PROJECT NAME:	690KW HEAT PUMP(R717)	 HEATING + COOLING WATER TO WATER HEAT PUMP-690KW (R717) REFRIGERATION PIPING AND INSTRUMENTATION DIAGRAM
PROJECT NUM:	2506-19	
REV:	9	
DATE:	25-SEP-25	